

1. The Core Conflict: Binary vs. Reality

Computers historically think in **Boolean Logic** (0 or 1). Humans think in **Degrees**.

The “Tall Person” Problem:

- **Crisp Logic (Boolean):**

- Define “Tall” as height $> 180\text{cm}$.
- Person A (181cm) = **Tall (1)**.
- Person B (179.9cm) = **Short (0)**.
- *Critique:* This abrupt change makes no sense in reality.

- **Fuzzy Logic:**

- Person A (181cm) = **0.95 Tall**.
- Person B (179.9cm) = **0.93 Tall**.
- *Result:* Smooth transition rather than a vertical cliff.